

Henrique Hoffmann Stefanutti
Independent Researcher
São Carlos, Brazil
henrique407@outlook.com

September 26, 2026

To the Editor-in-Chief
International Journal of Critical Infrastructure Protection

Dear Editor,

I am submitting the manuscript “From Cloud to Assembly Line: Modelling Cyberattack Paths in Automotive Manufacturing as Critical Infrastructure” for consideration as a research paper in the *International Journal of Critical Infrastructure Protection*.

Fit with the journal’s scope. The paper treats the automotive assembly plant, rather than the vehicle, as critical infrastructure. It models how a compromise propagates across corporate information technology, identity, cloud services, the industrial demilitarised zone and the control layers, and how a failure at one supplier cascades to an original equipment manufacturer through just-in-time coupling. That is the interdependency and cascading- failure problem the journal addresses, in a sector that regulators have recently brought within critical-infrastructure obligations. The work combines an engineering model with the regulatory language plants actually use, mapping every control to IEC 62443 system requirements and to the NIST Cybersecurity Framework.

Author. I am submitting as an independent researcher, with no institutional affiliation and no funding to declare.

Contribution. The paper contributes (i) a curated corpus of 20 publicly reported incidents in automotive manufacturing from 2017 to 2025, each reconstructed as an asset-level path with an explicit confidence label; (ii) a cross-plane attack-path model with an explicit probability semantics, instantiated as a reference plant of 12 zones, 48 assets and 91 attack steps; (iii) a quantitative analysis combining exact most-likely-path ranking with Monte Carlo campaign simulation; and (iv) a comparison against four baseline prioritisations, including one result that runs against our own model and is reported in full. The model, corpus and experiment pipeline are released openly, and every number in the paper is regenerated by a single reproducible run.

Originality and prior submission. The manuscript is original, has not been published elsewhere, has not been submitted to any other journal, and is not under consideration elsewhere.

Generative AI. Claude (Anthropic) was used as a writing and implementation assistant, under my direction, to draft and edit manuscript text and to implement and test the released software artefact. The research question, the model design, the choice of experiments and the interpretation of the results are mine, and I take full responsibility for the content. The manuscript carries the corresponding declaration.

Thank you for considering this submission.

Yours sincerely,
Henrique Hoffmann Stefanutti